1.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
–1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum(x_i-\bar{x}) =$ 0	$\sum(x_i-\bar{x})^2 =$ 10	$\sum(y_i-\bar{y}) =$ 0	$\sum(x_i-\bar{x})(y_i-\bar{y}) =$ 8

a. $\bar{x} = 1$, $\bar{y} = 2$

b. $b_2 = \dfrac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\Sigma(x_i-\bar{x})^2} = \dfrac{8}{10} = 0.8$

$b_1 = \bar{y} - b_2\bar{x} = 2 - 1 \times 0.8 = 1.2$

c. $\Sigma x_i^2 = 9+4+1+1 = 15$

$\Sigma x_i y_i = 12+4+3-1 = 18$

$\Sigma(x_i-\bar{x})^2 = 10$

$\Sigma x_i^2 - N\bar{x}^2 = 15 - 5\times 1^2 = 10 = \Sigma(x_i-\bar{x})^2$

$\Sigma(x_i-\bar{x})(y_i-\bar{y}) = 8$

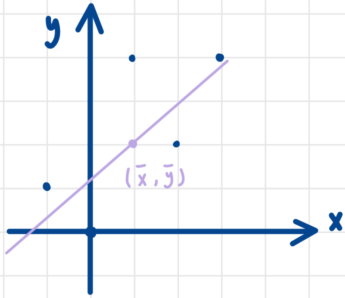
$\Sigma x_i y_i - N\bar{x}\bar{y} = 18 - 5\times 1\times 2 = 8 = \Sigma(x_i-\bar{x})(y_i-\bar{y})$

d. $\hat{y}_i = 1.2 + 0.8x_i$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
–1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$ 10	$\sum \hat{e}_i =$ 0	$\sum \hat{e}_i^2 =$ 3.6	$\sum x_i\hat{e}_i =$ 0

e.

f.



fitting line will always pass through $(\bar{x}, \bar{y})$

g. $\bar{y} = 2$, $b_1 + b_2\bar{x} = 1.2 + 0.8 \times 1 = 2 = \bar{y}$

h. $\bar{\hat{y}} = \sum_{i=1}^{5}(b_1 + b_2 x_i) / 5 = \frac{10}{5} = 2 = \bar{y}$

i. $\sum \hat{e}_i^2 = 0.4^2 + (-0.8)^2 + 1^2 + 0.6^2 + (-1.2)^2 = 3$

$\hat{\sigma}^2 = \frac{3.6}{5-2} = 1.2$

$se(b_2) = \sqrt{\frac{\hat{\sigma}^2}{\sum(x_i - \bar{x})^2}} = \sqrt{\frac{1.2}{10}} \approx 0.3464$

22. a.

```
Call:
lm(formula = totalscore ~ small, data = star5, subset = (aide ==
    0))

Residuals:
    Min      1Q  Median      3Q     Max
-283.04  -52.94   -6.94   44.96  321.06

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  918.043      1.667 550.664  < 2e-16 ***
small         13.899      2.447   5.681 1.44e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 74.65 on 3741 degrees of freedom
Multiple R-squared:  0.008553,	Adjusted R-squared:  0.008288
F-statistic: 32.27 on 1 and 3741 DF,  p-value: 1.441e-08
```

$\hat{y} = 918.043 + 13.899 \times small$

是小班 = 1，不是小班 = 0

$\Rightarrow$ 是小班制比不是的總分平均起來多 13.741 分

b.

```
Call:
lm(formula = readscore ~ small, data = star5, subset = (aide ==
    0))

Residuals:
     Min       1Q   Median       3Q      Max
-119.733  -21.733   -3.733   16.267  192.267

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 434.7327     0.7075 614.485  < 2e-16 ***
small         5.8191     1.0382   5.605 2.24e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 31.68 on 3741 degrees of freedom
Multiple R-squared:  0.008327,	Adjusted R-squared:  0.008062
F-statistic: 31.41 on 1 and 3741 DF,  p-value: 2.235e-08
```

$\hat{y} = 432.7327 + 5.8191 \times small$

$\Rightarrow$ 是小班制比不是的 read分平均起來多 5.8191 分

```
Call:
lm(formula = mathscore ~ small, data = star5, subset = (aide ==
    0))

Residuals:
    Min      1Q  Median      3Q     Max
-163.31  -34.31   -5.31   29.69  142.69

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  483.310      1.081 447.068  < 2e-16 ***
small          8.080      1.586   5.093  3.7e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 48.41 on 3741 degrees of freedom
Multiple R-squared:  0.006886,	Adjusted R-squared:  0.00662
F-statistic: 25.94 on 1 and 3741 DF,  p-value: 3.7e-07
```

$\hat{y} = 483.31 + 8.08 \times small$

$\Rightarrow$ 是小班制比不是的 math分平均起來多 8.08 分

c.

```
Call:
lm(formula = totalscore ~ aide, data = star5, subset = (small ==
    0))

Residuals:
    Min      1Q  Median      3Q     Max
-283.04  -50.04   -6.70   40.64  334.64

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 918.0429     1.6129 569.174   <2e-16 ***
aide          0.3139     2.2704   0.138     0.89
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 72.22 on 4046 degrees of freedom
Multiple R-squared:  4.725e-06,	Adjusted R-squared:  -0.0002424
F-statistic: 0.01912 on 1 and 4046 DF,  p-value: 0.89
```

$\hat{y} = 918.0429 - 0.3139$ aide

$\Rightarrow$ 有另一个隨班人員和總成績呈負相關，有的比沒有的少 0.3139

d.

y

```
Call:
lm(formula = readscore ~ aide, data = star5, subset = (small ==
    0))

Residuals:
     Min       1Q   Median       3Q      Max
-119.733  -21.438   -4.438   15.562  192.267

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 434.7327     0.6974 623.322   <2e-16 ***
aide          0.7054     0.9817   0.719    0.472
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 31.23 on 4046 degrees of freedom
Multiple R-squared:  0.0001276, Adjusted R-squared:  -0.0001195
F-statistic: 0.5163 on 1 and 4046 DF,  p-value: 0.4725
```

$\hat{y} = 437.7327 - 0.7054\,aide$

有隨班人員比沒有的 read 成績

低 0.7054

y

```
Call:
lm(formula = mathscore ~ aide, data = star5, subset = (small ==
    0))

Residuals:
     Min       1Q   Median       3Q      Max
-163.310  -33.919   -4.919   29.690  143.081

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 483.3102     1.0434 463.199   <2e-16 ***
aide         -0.3915     1.4687  -0.267     0.79
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 46.72 on 4046 degrees of freedom
Multiple R-squared:  1.756e-05, Adjusted R-squared:  -0.0002296
F-statistic: 0.07104 on 1 and 4046 DF,  p-value: 0.7898
```

$\hat{y} = 483.3102 - 0.3915\,aide$

有隨班人員比沒有的 math成績

低 0.3915

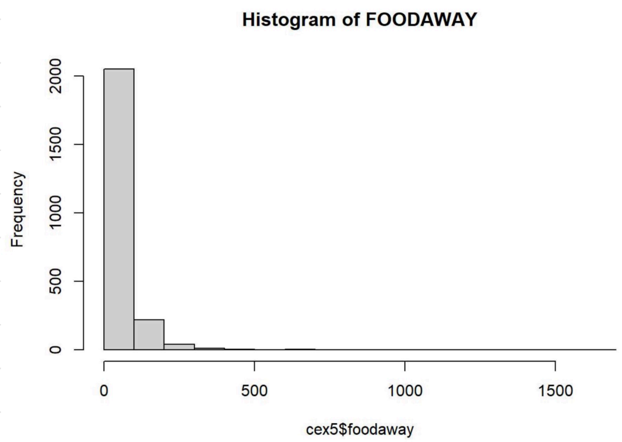
25.

a.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	14.12	33.15	51.45	69.44	1645.56

mean = 51.45 , medium = 33.15

25 th percentiles = 14.12 , 75 th percentiles = 69.44



b.

```
# advanced
summary(cex5$foodaway[cex5$advanced == 1])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   21.67   48.34   75.37   89.86 1179.00
```

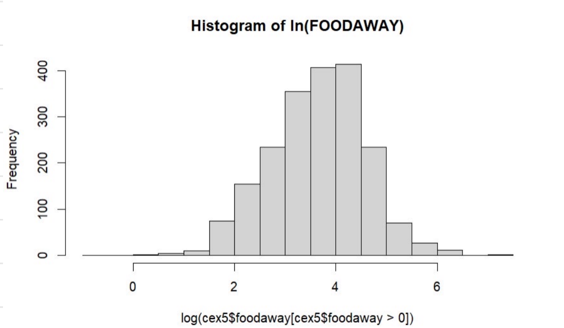
```
# college
summary(cex5$foodaway[cex5$college == 1])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   14.90   36.11   54.90   72.22 1645.56
```

```
# no advanced or college
summary(cex5$foodaway[cex5$advanced == 0 & cex5$college == 0])
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    9.63   25.02   38.32   55.97  437.78
```

c.



ln(foodway), foodway 觀測值數量不同，原因是數學.

f(x) = ln(x), only when x > 0 有定義, when x = 0, lnx → ∞

d.

```
## Call:
## lm(formula = log(foodaway) ~ income, data = cex5[cex5$foodaway
##     0, ])
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.3034 -0.6003  0.0654  0.6119  3.6262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.1364599  0.0420567   74.58   <2e-16 ***
## income      0.0069283  0.0004898   14.15   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.898 on 2003 degrees of freedom
## Multiple R-squared:  0.09083,    Adjusted R-squared:  0.09038
## F-statistic: 200.1 on 1 and 2003 DF,  p-value: < 2.2e-16
```

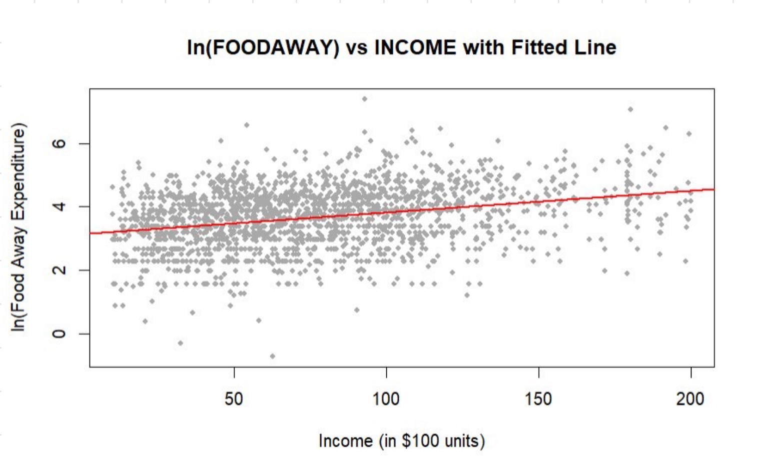
$\overset{y}{\ln(\text{foodway})} = 3.1365 + 0.0069\,\overset{x}{\text{income}}$

$\text{slope} = \frac{dy}{dx} = \beta_2 \exp(\beta_1 + \beta_2 x)$

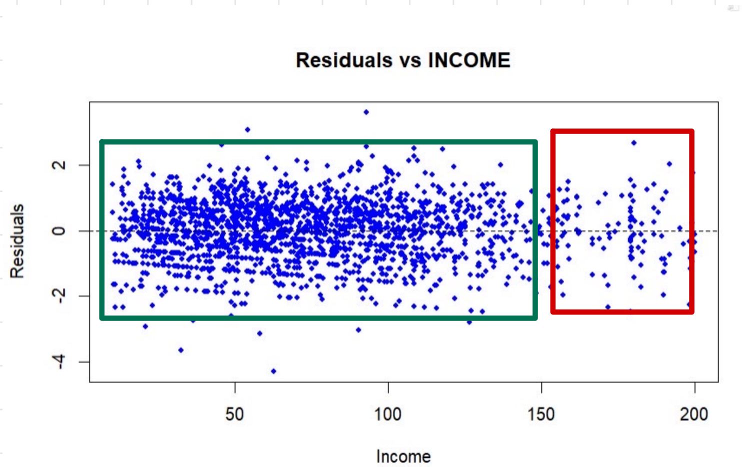
$= 0.062\,(3.1365 + 0.0069\,\text{income})$

income 增加 1 單位, 預期額外支出會增加 slope 倍

e.



f.



綠色区域及紅色区域有些不太均勻，但情況輕微
表示可能迴歸模型的假設会有問題